

# Legend

Formalized  
Result

Formalized  
Lemma

Conditional  
Bridge

Definition

Open /  
Pending

## Model Definitions

Utilities, policies, fairness

## Lemma 1 (App. C)

$I_{\min}^* > 0$

## Lemma 2 (App. C)

Item-fairness LP epigraph

## Proposition 2

Symmetric optimum exists

## Reduction Bridges

Original  $\leftrightarrow$  reduced model

## Theorem 3

First-half price monotonicity

## Problem 6

Opposing-type  
LP translation

## Lemma 4 (App. D)

Basic support +  
no-gap structure

## Lemma 5 (App. D)

Closed-form policy family

## Lemmas 6–11 (App. D)

Pivot, mono-  
tonicity, stitching

## Lemma 16 (App. E)

$q_j$  / midpoint algebra

## Problem 11 (App. E)

Three-type LP  
and equality form

## Lemma 12 (App. E)

Symmetric reduction into  $S'$

## Lemma 17 (App. E)

No right-half type-zero mass

## Lemma 13 Perturbations

$x$ -gap and  $z$ -gap certificates

## Lemma 13 (App. E)

Pivot support from  
equalized optimum

## Lemma 15 (App. E)

Pivot-one coordinates;  
full formula open

## Lemma 14 (App. E)

Uniqueness of optimum

## Theorem 4 No-Extremes

Equality-form Prob-  
lem 11 bridge

## Cold-Start Bounds

No-extreme  $\Rightarrow$  low utility

## No-Fairness Side

Estimated  $\gamma = 0$  benchmark

## True-Model Lower Bound

$U_{\min}^*(1, w) > 1/n$

## Theorem 4 Wrappers

Formalized from equality-  
form Problem 11

## Value-Vector Witness

Open  $v_2 < \varepsilon v_1/n$   
construction

## Theorem 4 (App. E)

Full paper theo-  
rem not closed yet